

Dr. Ashish Kumar

Professor of Physics

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Dr. Ashish Kumar received his M.Sc. in Physics from Kurukshetra University (2007) and his Ph.D. from the Department of Physics, Indian Institute of Technology (IIT) Delhi (2013). Following his doctoral degree, he joined the Inter-University Accelerator Centre (IUAC), New Delhi, as a Research Associate, and was subsequently awarded the prestigious DST-INSPIRE Faculty Fellowship (2015). During his tenure at IUAC (2013–2021), he was further honoured with the SERB Research Scientist designation (2020). In 2021, he joined the University of Petroleum and Energy Studies (UPES), Dehradun, before moving to the Central University of Jammu in 2023 as Associate Professor. He was promoted to Professor in the Department of Physics & Astronomical Sciences with effect from July 2026.

His research focuses on wide-bandgap semiconductor devices — including GaN, Ga₂O₃, and SiC-based HEMTs, Schottky barrier devices, and power electronics — alongside two-dimensional materials, thermoelectric energy generators, and ion beam modification of materials. SERB funded research develops printable and flexible nanocomposite thermoelectric materials for waste-heat energy recovery. With over 15 years of experience in ion-matter interactions and radiation effects in semiconductors, his group investigates defect physics and electronic transport using techniques including Deep Level Transient Spectroscopy (DLTS), photoluminescence, impedance spectroscopy, Raman spectroscopy, and Transmission Electron Microscopy (TEM). A strong interest in instrumentation has led him to design and develop several in-house characterization systems, including a Seebeck coefficient measurement setup, ZT and thermal conductivity measurement systems, a DLTS spectrometer, 1/f noise measurement facility, an optical dipstick cryostat, and low-cost vacuum ovens — significantly enhancing experimental capabilities at his laboratory.

Dr. Kumar has authored 91 publications in journals including Physical Review B, Journal of Applied Physics, Nuclear Instruments and Methods B, and RSC Advances, with an h-index of 27, an i10-index of 44, and over 1,850 citations on Google Scholar. He holds 3 patents and has delivered more than 30 invited talks at premier international and national venues, including the Materials Research Society Meeting (Boston, 2011), the International Conference on Nitride Semiconductors (Glasgow, 2011), DRIP XVII (Segovia, Spain, 2017), and the MCNMT Workshop at the International Centre for Theoretical Physics (ICTP), Trieste, Italy (2019). His research is funded by ISRO, DRDO, and IUAC, with cumulative external funding of approximately ₹1.45 crore. He has supervised three Ph.D. scholars to completion and currently mentors four doctoral researchers alongside M.Sc. and M.Tech. students.

In recognition of his contributions, Dr. Kumar was awarded the DST-INSPIRE Faculty Award (2015) and the SERB Research Scientist designation (2020) — among the highest honours conferred by the Department of Science and Technology. He was selected for the 69th Lindau Nobel Laureate Meeting (Germany, 2019), an invitation extended to fewer than 600 young scientists worldwide. His research group has earned notable recognition: a doctoral student received the RSC Advances Outstanding Paper Award (2024), the Best Thesis Award (2025), and the Best Poster Award at the DAE Solid State Physics Symposium (DAE-SSPS, 2025).

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